

EXCEL PRACTICAL ASSESSMENT: IF FUNCTION

Topic: IF, AND, OR, Nested IF & Advanced IF (Step-by-Step Practice)

General Instructions: Use the *Master Sample Table* for **Sections A through E**. Read each condition carefully before writing the formula. All outputs and logical criteria are **bolded** for clarity.

MASTER SAMPLE TABLE (USE FOR SECTIONS A–E)

Student	Marks	Attendance (%)	Project Marks	Sports
Amit	35	82	45	Yes
Riya	92	96	88	No
Sohan	58	70	62	Yes
Priya	76	85	74	No
Rahul	88	60	91	Yes
Neha	41	78	55	No
Arjun	67	91	81	Yes
Sneha	29	88	38	No

SECTION A – SIMPLE IF (5 QUESTIONS)

Requirement: Create a new column for each question using basic IF logic.

- Result Column:** If **Marks** are **40 or above**, display **"Pass"**. Otherwise display **"Fail"**.
- Attendance Status:** If **Attendance** is **75% or above**, display **"Regular"**. Otherwise display **"Irregular"**.
- Project Status:** If **Project Marks** are **50 or above**, display **"Project Passed"**. Otherwise display **"Project Failed"**.
- Merit Status:** If **Marks** are **80 or above**, display **"Merit Student"**. Otherwise display **"Normal Student"**.
- Sports Category:** If the **Sports** column contains **"Yes"**, display **"Sports Student"**. Otherwise display **"Non-Sports Student"**.

SECTION B – IF WITH AND (5 QUESTIONS)

Requirement: Use IF + AND to combine multiple mandatory conditions.

- Exam Eligibility:** A student is Eligible only if **Marks $\geq$ 40 AND Attendance $\geq$ 75%**. Otherwise display **"Not Eligible"**.
- Merit Certificate:** Give **"Merit Certificate"** only if **Marks $\geq$ 80 AND Project Marks $\geq$ 80**. Otherwise display **"No Certificate"**.
- Best Performer:** Display **"Best Performer"** only if **Marks $\geq$ 85 AND Attendance $\geq$ 90%**. Otherwise display **"Not Selected"**.
- Sports Award:** Display **"Sports Award Eligible"** only if **Sports = "Yes" AND Attendance $\geq$ 85%**. Otherwise display **"Not Eligible"**.
- Academic Excellence:** Display **"Academic Excellence"** only if **Marks $\geq$ 75, Attendance $\geq$ 80%, AND Project Marks $\geq$ 70**. Otherwise display **"No Award"**.

SECTION C – IF WITH OR (5 QUESTIONS)

Requirement: Use IF + OR where fulfilling any one condition suffices.

- Scholarship Eligibility:** Display **"Scholarship Eligible"** if **Marks $\geq$ 90 OR Project Marks $\geq$ 90**. Otherwise display **"Not Eligible"**.
- Needs Improvement:** Display **"Needs Improvement"** if **Marks below 40 OR Attendance below 75%**. Otherwise display **"Good Standing"**.
- Activity Selection:** Display **"Selected for Activity"** if **Sports = "Yes" OR Attendance $\geq$ 90%**. Otherwise display **"Not Selected"**.
- Special Recognition:** Display **"Special Recognition"** if **Marks $\geq$ 85 OR Attendance $\geq$ 95%**. Otherwise display **"No Recognition"**.
- Re-Test Required:** Display **"Re-Test Required"** if **Marks below 35 OR Project Marks below 40**. Otherwise display **"No Re-Test"**.

SECTION D – NESTED IF (5 QUESTIONS)

Requirement: Use sequential Nested IF functions to evaluate multi-tiered thresholds.

1. Grade Column: 90–100 = **A**, 75–89 = **B**, 60–74 = **C**, 40–59 = **D**, Below 40 = **Fail**.
2. Performance Column: Marks $\geq$ 90 = **Outstanding**, 80–89 = **Excellent**, 60–79 = **Good**, 40–59 = **Average**, Below 40 = **Poor**.
3. Division Column: Marks $\geq$ 80 = **First Division**, 60–79 = **Second Division**, 40–59 = **Third Division**, Below 40 = **Fail**.
4. Project Grade: Project Marks $\geq$ 85 = **A**, 70–84 = **B**, 50–69 = **C**, Below 50 = **Fail**.
5. Attendance Rating: Attendance $\geq$ 95 = **Excellent**, 85–94 = **Very Good**, 75–84 = **Good**, Below 75 = **Poor**.

SECTION E – COMBINATION (IF + AND + OR + NESTED IF) (5 QUESTIONS)

Requirement: Combine logical operators and nested conditions into composite formulas.

1. Final Exam Result: If Attendance is below 75%, display **"Not Eligible for Exam"**. Otherwise, if Marks $\geq$ 40 display **"Pass"**, else **"Fail"**.
2. Scholarship Category: **Gold Scholarship** if Marks $\geq$ 90 **AND** Attendance $\geq$ 90%; **Silver Scholarship** if Marks $\geq$ 80 **AND** Attendance $\geq$ 85%; **Bronze Scholarship** if Marks $\geq$ 70 **OR** Project Marks $\geq$ 85; Otherwise **No Scholarship**.
3. Prize Winner: Display **"Prize Winner"** if (Marks $\geq$ 85 **AND** Attendance $\geq$ 90%) **OR** Project Marks $\geq$ 90. Otherwise **"No Prize"**.
4. Student Status: **Star Student** if Marks $\geq$ 90, Attendance $\geq$ 90%, **AND** Project Marks $\geq$ 85; **Good Student** if Marks $\geq$ 75 **AND** Attendance $\geq$ 75%; **Average Student** if Marks $\geq$ 40; Otherwise **Needs Improvement**.
5. Overall Performance: **Excellent** if Marks $\geq$ 85 **AND** Project Marks $\geq$ 80 **AND** Attendance $\geq$ 85%; **Good** if Marks $\geq$ 70 **AND** Attendance $\geq$ 75%; **Average** if Marks $\geq$ 40; Otherwise **Poor**.

SECTION F – VERY ADVANCED LEVEL (REAL-WORLD COMPANY DATASET)

Requirement: Workplace analytics applying nested logic and compound operators to departmental KPIs.

SAMPLE TABLE – EMPLOYEE PERFORMANCE

Emp ID	Department	Sales (₹)	Attendance (%)	Target Achieved (%)	Customer Rating	Experience (Years)
E101	Sales	125000	98	115	4.9	6
E102	Sales	78000	90	96	4.5	3
E103	HR	45000	87	102	4.2	5
E104	IT	98000	72	110	4.8	7
E105	Support	52000	94	89	3.9	2
E106	Marketing	135000	96	120	4.7	8

VERY ADVANCED QUESTIONS (REAL OFFICE SCENARIOS)

1. Performance Bonus: **Bonus = ₹10,000** only if Sales $\geq$ ₹1,00,000, Attendance $\geq$ 90%, Target Achieved $\geq$ 100%, **AND** Customer Rating $\geq$ 4.5. Otherwise **Bonus = ₹0**.
2. Promotion Eligibility: **Eligible** only if Experience $\geq$ 5 years, Attendance $\geq$ 85%, Target Achieved $\geq$ 100%, **AND** Customer Rating $\geq$ 4.5. Otherwise **Not Eligible**.
3. Annual Performance Grade: **Platinum** if Sales $\geq$ ₹1,20,000 **AND** Target $\geq$ 110% **AND** Rating $\geq$ 4.8; **Gold** if Sales $\geq$ ₹90,000 **AND** Target $\geq$ 100% **AND** Rating $\geq$ 4.5; **Silver** if Sales $\geq$ ₹70,000 **AND** Target $\geq$ 90%; **Bronze** if Sales $\geq$ ₹50,000; Otherwise **Needs Improvement**.
4. Training Requirement: **Mandatory Training** if Attendance below 80% **OR** Target below 95% **OR** Customer Rating below 4.0. Otherwise **Training Not Required**.
5. Final Employee Status (Most Complex): Create one consolidated formula using **IF + AND + OR + Nested IF**:
 - **Star Performer**: Sales $\geq$ ₹1,20,000, Attendance $\geq$ 95%, Target $\geq$ 110%, Rating $\geq$ 4.8, Experience $\geq$ 5.
 - **High Performer**: Sales $\geq$ ₹90,000, Attendance $\geq$ 90%, Target $\geq$ 100%, Rating $\geq$ 4.5.
 - **Average Performer**: Sales $\geq$ ₹60,000 **AND** Attendance $\geq$ 80%.
 - **Needs Improvement**: Sales below ₹60,000 **OR** Target below 90%.
 - **On Probation**: Attendance below 75% regardless of other values.

Total Questions: 30 (Beginner – Real-World Advanced) | End of Assessment Paper